

Vulnerability Assessment to Identify Security Weaknesses

Tool: Nessus / OpenVAS

Theory

A vulnerability assessment is a systematic process of identifying and evaluating security weaknesses in a system. OpenVAS (Open Vulnerability Assessment Scanner) is a free, open-source scanner built into Kali Linux that scans a target for known vulnerabilities and rates them by severity using the CVSS score (0 to 10). It is used by security teams to find and fix weaknesses before attackers can exploit them.

Steps

1. Start Kali Linux and Metasploitable 2 in VirtualBox. Set both VMs to Host-Only network so they can communicate. Metasploitable is the intentionally vulnerable target machine.
 2. In Kali Linux terminal, install and set up OpenVAS. Run: `sudo apt update` then: `sudo apt install gvm -y`
 3. Initialise OpenVAS. Run: `sudo gvm-setup` — This downloads vulnerability feeds and sets up the database. Note the auto-generated admin password shown at the end of the output.
 4. Start OpenVAS services. Run: `sudo gvm-start` — Then open Firefox and go to: `https://127.0.0.1:9392` — Accept the SSL certificate warning and log in with username admin and the password from the previous step.
 5. Create a scan target. Go to Configuration > Targets and click the star icon to create a new target. Enter a name such as Metasploitable2 and set the host IP to 192.168.56.102. Save the target.
 6. Create a scan task. Go to Scans > Tasks and click the star icon. Enter a task name, select the target created in the previous step, set Scan Config to Full and Fast, and save the task.
 7. Start the scan by clicking the green Play button next to the task. The scan runs automatically and its progress is shown on screen. It may take 20 to 40 minutes to complete.
 8. Once the scan is complete, click on the task name and then click the date link under Last Report. The results page shows all discovered vulnerabilities sorted by severity: Critical, High, Medium, and Low.
 9. Click on any vulnerability to view its details including the CVE number, CVSS score, description, and recommended fix. Note down at least three critical or high severity findings for the report.
 10. To export the report, click Download Report at the top of the results page and select PDF or XML format. Save it to the Desktop.
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Conclusion

OpenVAS successfully scanned Metasploitable 2 and identified multiple vulnerabilities with severity ratings. The scan results demonstrate how automated vulnerability assessment tools help organisations find and prioritise security weaknesses before they are exploited.